Xiaotian Zhang

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EDUCATION

Northeastern University Seattle, USA

- **Degree:** Master of Science in Computer Science, **GPA:** 4.0/4.0 Sep 2024 - May 2026 (Expected)
- **Main Courses:** Programming Paradigms, Fundamentals of Artificial Intelligence, Database Management Systems, Software Engineering Fundamentals

Zhejiang University of Finance and Economics Hangzhou, China

- **Degree:** Bachelor of Engineering in Artificial Intelligence Sep 2020 - Jul 2024
- **Main Courses:** Probability and Statistics, Discrete Mathematics Generality, Design and Analysis of Algorithms, Internet and Web Design (HTML/CSS/JavaScript), Data Structure (C/C++), Data Visualization, Database Principles (SQL), Principles of Computer Organization, Game Theory and Its Applications, Object-Oriented Programming (Java), Advanced Python Programming, Embedded Interface Technology, Operating Systems, Deep Learning, Machine Learning, Computer Networks

RESEARCH EXPERIENCE

Worcester Polytechnic Institute May 2025 - Present

Position: Research Intern Advisor: A.P. Xiaozhong Liu

- Contributed to a patient adherence behavior research project, involving prompt engineering, LLM-based synthetic data generation, data preprocessing, and supervised fine-tuning (SFT) dataset creation.
- Generated patient-provider dialogues and adherence scenarios using LLMs to enhance training data, and trained several base models.
- Fine-tuned and evaluated the LLMs and their outputs to simulate and analyze patient adherence patterns.
- Built user interfaces that capture clinician feedback on model outputs and patient responses, creating a human-in-the-loop evaluation system for healthcare AI applications.
- Participated in the evaluation of the final model using a combination of methods.

Institute of Intelligent Machines, Chinese Academy of Sciences Jul 2023 - Aug 2023

Position: Research Intern Advisor: Dr. Zuchang Ma

- Participated in a project that predicted vascular health by collecting physiological data from diabetic foot patients at different disease stages and samples from healthy individuals.
- Assisted in building and processing datasets, and used machine learning and deep learning methods in classification prediction and cross-validation.
- Optimized models and processed a dataset of over 400 samples, achieving 85% accuracy in binary classification and 70% accuracy in ternary classification.
- Enhanced the dataset through correlation screening and feature selection, and improved the data quality with regularization and standardization methods.
- Adjusted parameters of machine learning models and fully connected layers of neural networks, significantly improving the prediction accuracy.

PROJECT EXPERIENCE

Database Application Development, Seattle, USA Jan 2025 - Apr 2025

Position: Team Leader

- Led a 6-member team to develop a GUI application based on MySQL and Java.
- Responsible for the creation and modification of the SQL database, focusing on the development of Java GUI and back-end database connections.
- Developed an application with basic CRUD (Create, Read, Update, Delete) functions, as well as features such as database system management and data visualization.
- Implemented calling the backend API, which could use machine learning to analyze users' social media data and recommend advertisements to the front-end accordingly.

IoT Application Development, Hangzhou, China Jan 2024 - May 2024

Position: Independent Developer

- Developed an IoT application based on Alibaba Cloud, focusing on C language programming for Jetson Nano and Warship development boards, and running Python programs on Jetson Nano via Alibaba Cloud and the back-end.
- Used Python's YOLO framework to develop a computer vision project, enabling the Warship development board to realize hard hat wearing detection via an external camera and the cloud communication to display results on the client side.
- Debugged and optimized the client-side C program running on Alibaba Cloud, designed a Python-based UI, realized remote control of components on the development board via the client such as access control switching and indicator light control, and enabled temperature sensing as well as data transmission to the cloud and client for display.

- Completed communication functions between the server, client, and development board to achieve seamless data transmission and command execution.

Online Insurance Assessment, Hangzhou, China Mar 2023 - Jun 2023

Position: Team Leader

- Led the team to conduct data cleaning, data mining, and model training/deployment for an online insurance assessment project based on Python.
- Responsible for team planning, requirement analysis, data processing and connection with the machine learning module.
- Collaborated on developing a Vue-based web front-end to create an assessment tool that determined insurance fraud risk with over 80% accuracy based on users' input.

Exploring the Role of AlphaFold2 and Deep Learning in Protein Structure Prediction (Taking Tau Tubulin Kinase 1 as an Example), Hangzhou, China Apr 2022 - Jun 2022

Position: Independent Developer

- Analyzed the AlphaFold2 protein structure prediction algorithm to show that AlphaFold2 improved accuracy by introducing an attention mechanism into the convolutional neural network of AlphaFold.
- Conducted protein structure prediction and comparison using Tau Tubulin Kinase 1 (TTBK1) as an example.
- Visualized the results with PyMOL, integrated them into the Anaconda virtual environment, compared predicted structures with actual protein structures, and demonstrated AlphaFold2's high accuracy in predicting basic protein structures.
- Studied the algorithm's impact on the ATP-binding site of TTBK1 to find the minimal deviation between the predicted and actual binding sites, which verified AlphaFold2's accuracy.

UNDERGRADUATE GRADUATION THESIS

Design and Implementation of a Construction Site Hard Hat Detection System Based on YOLOv5 and Alibaba Cloud Platform Jan 2024 – May 2024

Supervisor: Professor Gangyi Gao

- Developed a software system to meet construction site safety needs by realizing hard hat detection and operation control via Alibaba Cloud as the information terminal.
- Managed the cloud platform through gateway devices, edge computing units, and end terminals to ensure the stable operation of core functions.
- Utilized Alibaba Cloud SDK for platform connectivity, established reliable real-time communication using AMQP and MQTT protocols, and developed a user-friendly client interface using Python and PyQt5 for scheduling and monitoring of all system functionalities.
- Employed Jetson Nano for on-site image capture, deployed a pre-trained YOLOv5 model for object detection to identify whether personnel were wearing hard hat, and transmitted results and control instructions via gateways and virtual machines to update cloud and client status.

EXTRACURRICULAR ACTIVITIES

***Class Monitor,* Zhejiang University of Finance and Economics** Oct 2020 - Jun 2024

***Event Organizer,* College Art Troupe, Zhejiang University of Finance and Economics** Oct 2020 - Jun 2022

***Volunteer,* College Volunteer Activities, Zhejiang University of Finance and Economics** Apr 2021 - Jun 2022

SKILLS AND HOBBIES

Computer Skills: Python, Java, C, C++, HTML, CSS, JavaScript, SQL

Languages: Chinese (Native), English (Fluent)